

Lecture 1: Choice Theory & Consumer Theory

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September 2nd, 2025

Individual Decision-Making

- Microeconomic analysis starts from individual decisions.
- Rational choice: identify feasible options, pick the most preferred.
- Utility maximization represents this process.
- Historical roots:
 - Utilitarianism — “greatest good for the greatest number.”
 - Adam Smith — self-interest in economic systems.
- Strengths:
 - Policy relevance (welfare analysis).
 - Empirical support for comparative statics.
 - Wide scope: consumption, savings, education, crime, business.
 - Compact optimization framework.

Preferences

- Let X = set of possible choices.
- Weak preference: $x \succeq y \Leftrightarrow$ “ x at least as good as y ”.
- Strict preference: $x \succ y$ if $x \succeq y$ but not $y \succeq x$.
- Indifference: $x \sim y$ if $x \succeq y$ and $y \succeq x$.

Definition 1 (Completeness)

For all $x, y \in X$, either $x \succeq y$ or $y \succeq x$, or both.

Definition 2 (Transitivity)

If $x \succeq y$ and $y \succeq z$, then $x \succeq z$.

Example: Framing (Kahneman & Tversky, 1984)

- Buy stereo \$125, calculator \$15.
- \$5 discount on calculator at another store? Many say **Yes**.
- \$5 discount on stereo at another store? Many say **No**.
- \$5 discount on any of these two at another store? Indifferent.
- Same total savings, but framing affects choice $\Rightarrow$ violates transitivity.

$$C(B; \succeq) = \{x \in B \mid x \succeq y \ \forall y \in B\}$$

- $C(B; \succeq)$ = set of most-preferred items in B .
- Finite B : non-empty. Infinite B : may be empty.

Proposition 1

If $\succsim$ is complete and transitive:

- 1 For every finite non-empty B , $C(B; \succsim) \neq \emptyset$.
- 2 If $x, y \in A \cap B$, $x \in C(A; \succsim)$, $y \in C(B; \succsim)$, then $x \in C(B; \succsim)$ and $y \in C(A; \succsim)$.

Proof of Proposition 1 (Sketch)

- (1) Induction on $|B|$:
 - $|B| = 1$: $C(B) \neq \emptyset$.
 - Assume true for n , show for $n + 1$ using completeness and transitivity.
- (2) If $x \in C(A)$ and $y \in C(B)$, then $x \succeq y$ and $y \succeq z$ for all $z \in B$.
- By transitivity, $x \succeq z$ for all $z \in B$, hence $x \in C(B)$. Symmetric for $y \in C(A)$.

Choice and Revealed Preferences

- So far: derive choices from preferences.
- Empirics: observe choices, infer preferences.
- Question: Can observed choices be rationalized by complete, transitive preferences?

Choice Rule (as primitive)

Definition 3

Let $\mathcal{B}$ = collection of subsets of X . A choice rule is a function $C : \mathcal{B} \rightarrow \mathcal{B}$ with $C(B) \subseteq B$ for all $B \in \mathcal{B}$.

- Empirically, we try to learn C by observing choices.
- Question: when is C consistent with preference maximization?

Definition 4 (HARP)

A choice function $C : \mathcal{B} \rightarrow \mathcal{B}$ satisfies Houthakker's Axiom of Revealed Preference (HARP) if:

$$x, y \in A \cap B, x \in C(A), y \in C(B) \Rightarrow x \in C(B), y \in C(A).$$

Proposition 2

Statement: If $C : \mathcal{B} \rightarrow \mathcal{B}$ is non-empty, then there exists a complete and transitive $\succeq$ such that

$$C(\cdot) = C(\cdot; \succeq)$$

if and only if C satisfies HARP.

Proof of Proposition 2 (Sketch, Part I)

($\Rightarrow$)

- Suppose $C(\cdot) = C(\cdot; \succeq)$ for some complete, transitive $\succeq$.
- Then by Proposition 1, C satisfies HARP.

($\Leftarrow$) Assume C satisfies HARP.

- Define revealed preference relation:

$x \succeq_c y$ if there exists $A \subseteq X$, $y \in A$, $x \in C(A)$.

- Show $\succeq_c$ is complete:
 - For any x, y , let $A = \{x, y\}$.
 - If $C(A) = \{x\} \Rightarrow x \succeq_c y$.
 - If $C(A) = \{y\} \Rightarrow y \succeq_c x$.
 - If $C(A) = \{x, y\} \Rightarrow$ both $x \succeq_c y$ and $y \succeq_c x$.

Proof of Proposition 2 (Sketch, Part II)

Transitivity of $\succeq_c$:

- Suppose $x \succeq_c y$ and $y \succeq_c z$.
- Consider $C(\{x, y, z\})$.
- By HARP:
 - If $y \in C(\{x, y, z\})$, then $x \in C(\{x, y, z\})$.
 - If $z \in C(\{x, y, z\})$, then $y \in C(\{x, y, z\})$, hence $x \in C(\{x, y, z\})$.
- Thus $x \succeq_c z$.

Equivalence of C :

- If $x \in C(A)$, then $x \succeq_c y$ for all $y \in A$, so $x \in C(A; \succeq_c)$.
- Conversely, if $x \in C(A; \succeq_c)$, then by HARP $x \in C(A)$.

- HARP requires observing $C(A)$ for all $A \subseteq X$.
- Real data: limited (e.g. only budget sets).
- Weaker condition: **WARP** (Weak Axiom of Revealed Preference).
- Further results in textbooks (see, e.g., MWG).

Utility Representation

- Preferences are abstract $\Rightarrow$ utility gives numerical ranking.
- $u : X \rightarrow \mathbb{R}$ represents $\succeq$ if

$$x \succeq y \Leftrightarrow u(x) \geq u(y).$$

- Then choice: $C(B; \succeq) = \arg \max_{y \in B} u(y)$.

Proposition 3

If X is finite, any complete and transitive $\succeq$ can be represented by some u .

Proof (sketch): Induction on $|X|$: assign top choice utility $n + 1$, recurse on remaining n elements.

Infinite Sets and Continuity

- Not all $\succeq$ are representable by u .
- Example: **Lexicographic preferences**: prioritize x_1 before x_2 .
- No real-valued utility function exists.

Definition 6 (Continuity)

$\succeq$ continuous if $(x_n, y_n) \rightarrow (x, y)$ and $x_n \succeq y_n \forall n \Rightarrow x \succeq y$.

Proposition 4

If $X \subseteq \mathbb{R}^n$ and $\succeq$ is complete, transitive, continuous, then $\succeq$ has a continuous utility representation.

Proof Idea of Proposition 4

- Assume strictly monotone $\succeq$ on $X = \mathbb{R}_+^n$.
- For each x , find $\alpha(x)$ with $\alpha e \sim x$ ($e = (1, \dots, 1)$).
- Define $u(x) = \alpha(x)$.
- Ensures u represents $\succeq$; continuity $\Rightarrow u$ continuous.
- Intuition: indifference curve crosses 45° line.
- Utility not unique: any increasing transform $v(u(\cdot))$ also represents $\succeq$.
- No interpersonal comparisons of utility levels.

Variants of Utility

- **Interpersonal comparisons:** Veil-of-ignorance arguments.
- **Just noticeable differences:** $x \succ y \Leftrightarrow u(x) \geq u(y) + 1$.
- Captures insensitivity to small changes, stronger response to large ones.

Why Restrictions?

- General preference assumptions are too broad.
- Restrictions make analysis tractable and predictions testable.
- Common restrictions:
 - Monotonicity & Local Non-satiation
 - Convexity
 - Separability
 - Quasi-linearity (no wealth effects)

Basic Restrictions

Definition 7 (Monotonicity)

$x \geq y \Rightarrow x \succeq y$.

Definition 8 (Local Non-satiation)

For any y , $\varepsilon > 0$, $\exists x$ near y with $x \succ y$.

Definition 9 (Convexity)

If $x \succeq y$ and $x' \succeq y$, then $tx + (1 - t)x' \succeq y$ for $t \in (0, 1)$.

Proposition 5

If u represents $\succeq$:

- ① $\succeq$ monotone $\Leftrightarrow u$ non-decreasing.
- ② $\succeq$ locally non-satiated $\Leftrightarrow u$ has no local maxima.
- ③ $\succeq$ convex $\Leftrightarrow u$ quasi-concave.

Quasi-concave Functions: Mathematical Definition

Upper contour characterization

Let $f : C \rightarrow \mathbb{R}$ with $C \subseteq \mathbb{R}^n$ convex.

f is **quasi-concave** iff for every $\alpha \in \mathbb{R}$, the set

$$U_\alpha = \{x \in C \mid f(x) \geq \alpha\}$$

is convex.

Pointwise characterization

For all $x, y \in C$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}.$$

- Strict quasi-concavity: inequality is strict when $x \neq y$ and $f(x) \neq f(y)$.
- Link: convex preferences $\iff$ quasi-concave utility.

Separability

- Sometimes choices can be partitioned: $x \in X, y \in Y$.
- x choice does not depend on y .

Definition 10

$x^*(S, y)$ independent of y if $\forall y_0, y_1 \in Y, x^*(S, y_0) = x^*(S, y_1)$.

Proposition 6

If $\succeq$ has utility $u(x, y)$, then x^* independent of $y \Leftrightarrow$

$$u(x, y) = U(v(x), y),$$

where $v : X \rightarrow \mathbb{R}$ and U increasing in first argument.

Quasi-linear Preferences (No Wealth Effects)

- Choice space: $X = \mathbb{R}_+ \times Y$, with $a \in \mathbb{R}_+ = \text{"money"}$.
- Assumptions:
 - 1 Good 1 is valuable: $(a, y) \succeq (a', y)$ iff $a \geq a'$.
 - 2 Compensation possible: $\forall y, \exists t$ s.t. $(0, y) \prec (t, \bar{y})$.
 - 3 No wealth effects: if $(a, y) \succeq (a', y')$, then $(a + t, y) \succeq (a' + t, y')$ for all t .

Proposition 7

These conditions $\Leftrightarrow \exists v : Y \rightarrow \mathbb{R}$ such that

$$(a, y) \succeq (a', y') \Leftrightarrow a + v(y) \geq a' + v(y').$$

Utility: $u(a, y) = a + v(y)$.

Behavioral Criticisms of Rational Choice

- Rational choice is powerful. Also limited.
- Key critique: choices depend on context.
- People use heuristics. Cognition is limited.
- Experiments show systematic deviations.
- Policy analysis implications.

Context Dependence & Framing

- Presentation matters. Small cues shift choices.
- Temptations alter reference points. E.g., dessert menu.
- Adding irrelevant options can change selections.
- If preferences shift with context, portability is reduced.
- Harder to infer behavior across environments.

Heuristics & Bounded Rationality

- People cannot optimize everywhere.
- Use rules of thumb. Learn from past success.
- Standard models may fit in familiar settings.
- In unfamiliar settings, deviations are large.
- Modeling challenge: which heuristics, when?

Evidence I: Anchoring & Coherent Arbitrariness

- Ariely–Loewenstein–Prelec (2003, QJE).
- Students valued goods after seeing random SSN digits.
- Valuations correlate with the anchor.
- After anchoring, responses to price changes look coherent.
- Lesson: initial values can be arbitrary, then internally consistent.

Evidence II: Defaults & Opt-In/Opt-Out

- Madrian & Shea (2001, QJE): 401(k) enrollment.
- Opt-in vs opt-out yields large participation gaps.
- Organ donation: U.S. opt-in vs Spain opt-out.
- Defaults shape outcomes, even with large stakes.
- Hard to reconcile with strict optimization.

Modeling Responses

- Allow preferences to depend on context.
- Incorporate temptation, cues, and framing.
- Bounded rationality: partial optimization, mistakes.
- Learning models: adaptive rules, habit formation.
- Trade-off: realism vs tractability and predictive power.

Implications for Policy Analysis

- Rational choice simplifies welfare analysis.
- Assumes stable, outcome-focused preferences.
- Behavioral critiques question these assumptions.
- Alternatives must still guide clear policy choices.
- Open question: robust, powerful, realistic models.